

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_best_single_fixed_heuristic, phase8_full_solver_evolution, phase8_llm_evolved_adaptive_controller.

Run Metadata

- run_name: run_20260520_185510_f
- started_at_local: 2026-05-20 18:55:10
- finished_at_local: 2026-05-20 19:24:08
- duration_hhmm: 00:29
- duration_seconds: 1738.022
- seed_offset: 5000
- replicate_label: f
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1451.927243	0.07226	0.007284	0.03122	0.0	0.0	True
phase8_random_portfolio	random_portfolio	none	0.187933	0.115673	1486.783929	0.192445	0.04395	0.096996	0.0	0.0	False
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1452.1786	0.072273	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1523.144029	0.075804	0.007284	0.03122	0.0	0.0	False
phase8_llm_static_selector	static_selector	none	0.214311	0.142051	465.010557	0.214311	0.052514	0.112124	0.0	0.62	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.201598	0.129338	299.300657	0.201598	0.04628	0.103503	0.0	0.62	True
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.246315	0.174055	562.769286	0.246315	0.001913	0.091956	0.12068	0.62	False
phase8_full_solver_evolution	full_solver	none	0.110275	0.038015	387.225186	0.110275	0.000881	0.041184	0.771891	0.76	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1451.927243 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.187933, selector regret 0.115673, runtime-adjusted gap 0.192445.
- Family transfer 0.04395 and combined transfer 0.096996.
- Runtime 1486.783929 ms, runtime inflation 0.024007, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.072273.
- Family transfer 8.9e-05 and combined transfer 0.026678.

- Runtime 1452.1786 ms, runtime inflation 0.000173, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.075804.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1523.144029 ms, runtime inflation 0.04905, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.052514 and combined transfer 0.112124.
- Runtime 465.010557 ms, runtime inflation -0.679729, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cheapest_insertion_two_opt:limited_three_opt:edge_preserving_restart:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": 2, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "edge_preserving_restart", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive frozen-portfolio selector using structure descriptors; chooses an interpretable heuristic per regime and provides bounded schedule/tuning parameters for later-stage refinement.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "limited_three_opt", "low_improvement_threshold": 0.05, "name": "tsp_ihx_frozen_portfolio_epoch1_static_rules", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3, "temperature_scale": 1.05, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "limited_three_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.201598, selector regret 0.129338, runtime-adjusted gap 0.201598.
- Family transfer 0.04628 and combined transfer 0.103503.
- Runtime 299.300657 ms, runtime inflation -0.79386, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:annealed_multi_start:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": 2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "annealed_multi_start", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive controller selecting from a frozen heuristic portfolio with bounded online schedule tuning based on stagnation, improvement rate, and perturbation failures.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "cheapest_insertion_two_opt", "low_improvement_threshold": 0.06, "name": "det_tsp_hh_oracle_style_portfolio_adaptive_v1", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "annealed_multi_start",

"stagnation_threshold": 3, "temperature_scale": 1.15, "time_budget_trigger": 0.7, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"]}

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.246315, selector regret 0.174055, runtime-adjusted gap 0.246315.
- Family transfer 0.001913 and combined transfer 0.091956.
- Runtime 562.769286 ms, runtime inflation -0.612398, mean novelty 0.12068, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cheapest_insertion_two_opt:farthest_insertion_two_opt:nearest_neighbor_multistart:stag4:cand-1:restart2.
- Controller rule summary: {"acceptance_bias": 0.008, "candidate_limit_offset": -1, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive hyper-heuristic controller selecting one frozen TSP heuristic and tuning bounded schedule knobs from online stability signals.", "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.12, "name": "interpretable_adaptive_tsp_hh_controller_v2", "nearest_neighbor_trap_heuristic": "cluster_first_local_search", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 2, "stagnation_switch_heuristic": "nearest_neighbor_multistart", "stagnation_threshold": 4, "temperature_scale": 1.25, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.110275, selector regret 0.038015, runtime-adjusted gap 0.110275.
- Family transfer 0.000881 and combined transfer 0.041184.
- Runtime 387.225186 ms, runtime inflation -0.733303, mean novelty 0.771891, and complexity 0.76.
- Pareto efficient: True.

Judge Appendix

Conservative Interpretation of TSP Adaptive Heuristic Portfolio Suite Results

Prioritized Metrics Summary

Condition	Held-out TSPLIB Gap	Selector Regret	Runtime-Adjusted Gap	TSPLIB Runtime (ms)	Runtime Inflation	Transfer Gap	Pareto Efficient	Comments
phase8_best_single_fixed_heuristic (Best TSPLIB)	**0.07226**	0.0	0.07226	1452	0.0	0.0312	Yes	Best fixed heuristic, no runtime inflation, solid baseline
phase8_oracle_selector (Best Transfer)	0.07226	0.0	0.07227	1452	~0	**0.0267**	No	Oracle upper bound, not deployable, validates best fixed result
phase8_llm_evolved_adaptive_controller (Pareto)	0.2016	0.1293	0.2016	299	-0.79	0.1035	Yes	Adaptive, much faster runtime, significant regret, runtime gain

Condition	Held-out TSPLIB Gap	Selector Regret	Runtime-Adjusted Gap	TSPLIB Runtime (ms)	Runtime Inflation	Transfer Gap	Pareto Efficient	Comments
phase8_full_solver_evolution (Pareto)	0.1103	0.0380	0.1103	387	-0.73	0.0412	Yes	Evolved full solver, moderate gap and regret, strong runtime gain
phase8_llm_static_selector	0.2143	0.1421	0.2143	465	-0.68	0.1121	No	Static selector, worse gap and regret despite runtime reduction
phase8_llm_evolved_controller_diversity_failure_replay	0.2463	0.1740	0.2463	563	-0.61	0.0920	No	Adaptive with replay, highest gap and regret, moderate runtime gain
phase8_random_portfolio	0.1879	0.1157	0.1924	1487	0.02	0.0970	No	Random baseline, poor gap and regret, no runtime improvement

Key Insights

1. **Held-out TSPLIB Gap (Primary Endpoint)**

- The **best single fixed heuristic (farthest_insertion_two_opt)** achieves the lowest held-out TSPLIB gap (0.07226), tied with the oracle selector (upper bound).
- Adaptive selectors and evolved controllers have significantly higher gaps (0.1103 to 0.2463), indicating no definitive improvement in solution quality yet.

2. **Selector Regret vs Oracle Portfolio**

- Oracle selector regret is zero by definition (upper bound).
- The best adaptive controller (phase8_full_solver_evolution) shows moderate regret (0.038), while other adaptive methods regret is larger (0.13-0.17), indicating imperfect instance adaptation.
- The fixed heuristic selector has zero regret but no adaptability.

3. **Runtime-Adjusted Gap & Runtime Inflation**

- The best fixed heuristic has zero runtime inflation but longer runtime (~1452 ms).
- Adaptive controllers and evolved solvers offer large negative runtime inflation (-0.73 to -0.79), i.e., **significant runtime speedups**.
- However, these runtime reductions come with increased performance gap, reflecting a tradeoff between speed and quality.
- The static selector also reduces runtime but at cost of poorer gaps and higher regret.

4. **Pareto Efficiency**

- Three conditions are Pareto-efficient: the fixed best heuristic, the evolved full solver, and the evolved adaptive controller.
- These represent meaningful tradeoffs along the quality-runtime spectrum.

- Adaptive controllers provide faster runtime with moderate quality loss; best fixed heuristic optimal for highest solution quality without runtime gain.

5. ****Cross-family Transfer****

- Transfer gaps are smaller for the best fixed heuristic (0.0312) and evolved full solver (0.0412) than adaptive controllers (0.091-0.103).
- Oracle selector lowest transfer gap (0.027), as expected.
- This suggests adaptive controllers may have less robust transfer/generalization and adaptation efficiency can be improved.

6. ****Instance-Adaptive Control Interpretability****

- Adaptive controllers use interpretable frozen heuristic portfolios with rules tuned on stagnation and improvement signals.
- However, these do not yet approach the oracle in gap or regret, limiting claims of near-oracle interpretability without runtime inflation.
- Static selector simpler but has poorer tradeoffs and should not be preferred for deployment.

Summary Statement

The best single fixed heuristic ("farthest_insertion_two_opt") remains the strongest baseline with the lowest held-out TSPLIB gap and zero selector regret, serving as a solid upper baseline for practical deployment without runtime cost. The oracle selector corroborates this performance as an unattainable upper bound.

Adaptive hyper-heuristic controllers and evolved solvers provide significant runtime reductions (up to ~79% faster) and represent meaningful Pareto tradeoffs, though with increased solution quality gaps and selector regret. Among these, the evolved full solver displays moderate regret with fair, lower gap and runtime gains, highlighting the potential for interpretable instance-adaptive control within a bounded schedule tuning framework. Cross-family transfer remains a challenge, with adaptive selectors showing higher transfer gaps than fixed heuristics.

Given the current results, conservative claims focus on effective **interpretable instance-adaptive portfolio control** enabling runtime-quality tradeoffs, rather than discovery of better heuristics or near-oracle adaptive performance. Further improvements should target reducing selector regret and closing the quality gap while preserving runtime benefits.